Local Global Games and Network Common Learning

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Abstract

We study global games in which agents coordinate locally, with their social network neighbors, contingent on a favorable state. Before acting, agents learn the private signals of all agents within network distance r. As r grows, every agent learns the state, but efficient coordination depends on higher-order beliefs, which are shaped by the geometry of the network. We introduce network common learning, a network analogue of common learning (Cripps et al., 2008), and show that it is attained when neighboring agents' observations differ by many signals, as on the two-dimensional grid, but fails on networks with informational bottlenecks, such as the line, where only the safe action survives in equilibrium.

1 Introduction

Global games are a successful and well-studied model of a number of important phenomena, including bank runs, protests and more (Carlsson and van Damme, 1993; Morris and Shin, 2003; Goldstein and Pauzner, 2005; Angeletos et al., 2007). The basic model is of a group of agents that need to act in unison to achieve a goal, and furthermore can only succeed when the state of nature is favorable. The important insight they provide is that even when agents have very precise information about the state, coordination can fail because of higher-order uncertainty (Rubinstein, 1989; Carlsson and van Damme, 1993; Morris and Shin, 2003). The study of global games has spurred a literature within epistemic game theory, dedicated to understanding how beliefs affect coordination. A particularly interesting concept is the idea of common learning (Cripps, Ely, Mailath, and Samuelson, 2008), which captures a convergence not just to knowing the state, but to commonly knowing it (see also Steiner and Stewart, 2011; Cripps, Ely, Mailath, and Samuelson, 2013; Frick, Iijima, and Ishii, 2023).

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Our setting features a classic global game. There is a state that is either high or low, and each agent has to choose between a safe and a risky action. When the state is low, the safe action is optimal. When the state is high, it is better to take the risky action, but only if others take it too. We introduce a network structure to this game, so that agents do not need to coordinate with all others, but only locally, with their social network neighbors; accordingly, we call this game a *local global game*. This changes the higher-belief structure required for coordination. Instead of the common q-belief that is needed for coordination on a complete network, a general network requires a weaker condition over iterated beliefs, which we call network q-beliefs, and which we show is equivalent to the existence of an efficient equilibrium. This is similar to the conditions explored by Morris, Rob, and Shin (1995) and Kets (2011).

Our network plays an additional important role, as a conduit of information. Initially, agents are endowed with conditionally independent private signals about the state. Before choosing their actions, the agents participate in a communication stage, in which they share their signals with their neighbors, their neighbors' neighbors and so on, up to some graph distance r, which we call the observation radius. For simplicity, we model this communication phase non-strategically, and assume that players simply reveal their private information. Equivalently, we can think of these private signals as locally-public signals, i.e., signals that are observable by agents up to distance r.

For large r, each agent learns the state with high confidence. Moreover, neighboring agents observe information that is highly overlapping, seemingly allowing their beliefs to align more easily. But does individual learning and the overlap of information allow agents to resolve strategic uncertainty and coordinate efficiently?

We show that the answer to this question depends sharply on the geometry of the network. Consider two stylized examples: the infinite line and the infinite two-dimensional grid. On both networks, increasing the observation radius allows every agent to learn the state with arbitrarily high confidence. Nevertheless, the equilibrium implications are different. On the line, when coordination requires sufficiently high confidence, the safe action is the only action taken in equilibrium, no matter how large the radius r is. On the two-dimensional grid, by contrast, when the radius is large there exist efficient equilibria in which the agents, with high probability, coordinate and take the risky action in the high state.

Why does the additional dimension reverse the equilibrium prediction? The key is not how much information each agent observes but how the available information changes across agents. Two neighboring agents observe largely overlapping neighborhoods, yet each also sees some signals that the other does not. We refer to signals observed by one agent but not by his neighbor as his *information boundary*. On the line, this region contains only one signal, regardless of the size of the observation radius. On the two-dimensional grid, the information boundary grows proportionally with the observation radius.

This difference determines the feasibility of coordination. When the size of the

information boundary remains bounded, there is no concentration of measure, and the uncertainty about a neighbor's information stays significant, even when r is large. This concern propagates from neighbor to neighbor, eventually ruling out risky coordination. On the other hand, when the information boundary is large, the unobserved signals become predictable, because of concentration of measure. Strategic uncertainty diminishes as the radius grows, allowing confidence in risky play to sustain itself across the network.

To construct efficient equilibria, we study what we call *network q-belief* (see also Morris, Rob, and Shin, 1995; Kets, 2011). This notion is defined recursively. At the first level, an agent is required to assign probability at least q to the state being high. At each subsequent level, he needs to assign probability at least q to the joint event that the state is high and that all his neighbors assign probability q to the events defined at the preceding level. We show that the limit of this iterated belief satisfies the same fixed-point condition as the agents' best responses in equilibrium. Namely, it yields a collection of *self-sustaining events*: for each agent, whenever his corresponding event in the collection happens, he assigns probability at least q to the joint event that the state is favorable and that every neighbor's corresponding event also happens. Thus, there is an equilibrium such that, when this event happens, the agent takes the risky action. In analogy to the idea of common learning (Cripps, Ely, Mailath, and Samuelson, 2008), we define *network common learning*, which holds if, for all q, network q-belief is obtained with probability approaching one as r tends to infinity.

Our first result is an impossibility theorem (Theorem 1). Suppose that the network contains a "bottleneck": a path along which moving from one agent to the next introduces only a bounded number of signals not observed by the previous agent; this, for example, holds for the one-dimensional grid. This implies that the higher-order uncertainty generated by these non-overlapping signals remains significant regardless of r, and is not mitigated by concentration of measure. Consequently, when q is high enough, the collection of events that can be believed with probability q shrinks as we iterate over higher-order beliefs and move along the path, diminishing to the empty set if the path is long enough. Consequently, network q-belief never occurs for any agent, and the safe action is the unique equilibrium prediction.

In contrast to this impossibility result, we next establish sufficient conditions for network common learning. We show that these conditions are satisfied by two broad classes of networks: "convex" subgraphs of grids (Theorem 2), and bounded-degree trees (Theorem 3), when the growth of the agents' observation neighborhoods is sufficiently large.

The proof relies on the structure of the information that is relevant for higher order beliefs. To capture this structure, we introduce *test regions*: these are regions in the social network graph that are finite intersections of the agents' observation neighborhoods. For both classes, the geometry of their test regions allows us to control uncertainty that accumulates across neighboring agents. Sufficiently fast growth of neighborhoods then ensures that this uncertainty is relatively small. This allows us

to construct self-sustaining events that yield efficient equilibria. The self-sustaining events require that in each test region the share of favorable signals is large. That is, to coordinate an agent needs to not just see many favorable signals (yielding individual learning) but see many favorable signals in each test region.

Omer: Rethink next paragraph. Maybe unite it with the previous paragraph?

We then give complementary sufficient conditions that apply to a more general class of networks. The key geometric property is what we call *large information boundaries* (Definition 4). Large information boundaries are attained when every large test region that is observed by some agent i but is not seen by i's neighbor j is large (or empty). The term "large" is used informally here, but is defined formally below. Together with the restriction on the number of test regions and on the relative sizes of observation neighborhoods, this condition also implies network common learning (Theorem 4).

Our paper identifies a new channel through which network geometry affects coordination. Network geometry matters not only because it determines who interacts with whom and how information is diffused, but also because it determines how much uncertainty remains when agents reason about the actions of their neighbors. When the uncertainty about others' information becomes small relative to the amount of information agents observe, confidence in others' actions can be sustained. Otherwise, this uncertainty may propagate over long distances, even though every individual agent is highly informed about the state.

1.1 Related literature

Our model follows the literature on global games (Carlsson and van Damme, 1993; Morris and Shin, 2003). The basic lesson of this literature is that arbitrarily precise private information need not induce coordination because equilibrium behavior also hinges on higher-order beliefs (Rubinstein, 1989). We inherit this lesson but modify the object of higher-order beliefs. In the canonical model, an agent needs to coordinate with the entire population, and this imposes conditions on beliefs that involve every agent at every epistemic order. In our model, an agent's payoff only depends on the actions of his neighbors, and so the relevant object is network q-belief, which chains beliefs along the edges of the network. Despite this difference from the classic literature, our impossibility result is in the same family as the coordinated attack problem of Halpern and Moses (1984): the argument in the proof of Theorem 1 is a spatial version of the temporal mechanism governing the electronic mail game of Rubinstein (1989), with graph distance along a bottleneck path playing the role of the number of messages exchanged.

Network q-belief is a network analogue of common p-belief, as defined by Monderer and Samet (1989), who show that common p-belief admits a fixed-point characterization; we derive a network analogue of their fixed-point property. Common p-belief is also central to the analysis of Kajii and Morris (1997), who show that an equilibrium

is robust to incomplete information when the event on which the game is common knowledge can be commonly p-believed, and whose contagion arguments are echoed in our impossibility result. Kets (2011) is closest to us in adapting the belief operator to the network structure. Beyond uncertainty about the state, Kets (2011) also considers uncertainty about the network structure itself, and applies these techniques to the study of anonymous network games.

Morris (2000) studies how local interactions determine whether behavior can spread contagiously through a network. In his complete information setting, agents repeatedly best respond to the actions of their neighbors. He shows that an action adopted by a finite group of agents can be spread through the entire network only when there is no cohesive group that interacts mostly within the group and resists the outside contagion. Our analysis shares the emphasized propagation and recursive structure but the object being propagated is different. In Morris (2000), an action diffuses through the network. In contrast, we study how uncertainty propagates through higher-order beliefs and the network structure. Network geometry therefore matters not only through who interacts with whom, but also through how information changes across neighboring positions. Related work on local interaction dynamics and geometry includes Ellison (1993), Blume (1993) and Oyama and Takahashi (2015).

Cripps, Ely, Mailath, and Samuelson (2008) study agents who accumulate private signals over time and ask whether this leads to common p-belief. They show that when the number of signal realizations is finite, individual learning implies common learning; see also Frick, Iijima, and Ishii (2023). Our observation radius r plays the role of time in their model: as the radius r grows, the information of each agent becomes arbitrarily precise and the question of interest is whether higher-order beliefs also grow. In our model, individual learning does not always imply common learning, even with a finite number of signals; it is the geometry of the graph that decides.

Chwe (2000) also studies coordination in which a network is used to communicate, and asks which networks are minimally sufficient for coordination. In his model communication is used to build exact common knowledge within the set of participants, so the binding constraint is whether a network has the right links.

A recent paper by Hutchcroft, Rospuskova, and Tamuz (2026) also studies coordination on social networks using local communication, but in a crucially different setting, resulting in very different conclusions. In that paper, there is no global state, and agents play a pure coordination game (e.g., choosing which side of the road to drive on). Thus, the challenge is how to use communication to break ties between a priori equivalent actions. In that setting, coordinating on the line graph is possible, and in fact higher efficiency can be achieved on the line than on the two-dimensional grid. Amenability, the geometric property enabling coordination in that paper, is related to having small information boundaries, while in our setting large information boundaries are a sufficient condition for coordination.

2 Model

Let N be a set of agents, connected through a social network $G = (N, E)$. An edge $(i, j) \in E$ indicates that agents i and j are neighbors. We assume that the social network is undirected, so that $(i, j) \in E$ if and only if $(j, i) \in E$. For convenience, $(i, i) \notin E$. For each $i \in N$, let $N_i = \{j \in N : (i, j) \in E\}$ denote the set of i's neighbors. We assume that every agent has finitely many neighbors, i.e. $|N_i| < \infty$ for every $i \in N$. A path from i to j is a finite sequence of agents beginning at i and ending at j, such that every two consecutive agents are neighbors. We assume that G is connected, i.e., that there is a path connecting each pair of agents. This is without loss of generality, since otherwise one can consider each connected component in isolation. We will be interested in the limit of very large networks, and so for simplicity we will assume that the number of agents is countably infinite. **Omer:** comment on finite networks.

The distance between i and j is the length of the shortest path between them. Each agent observes all agents at graph distance at most r, where $r \geq 0$. We call r the observation radius. We also define $B_r(i) \subseteq N$ as the set of agents at distance at most r from i. We refer to this set as r-neighborhood of i, or as the observation neighborhood of agent i.

We consider a coordination game played on the network G. The payoff of an agent depends on the underlying state, his own action and the actions of his neighbors. The state is binary: $\theta \in \{H, L\}$, with H and L denoting the high and low states respectively. Each agent chooses between a safe action S and a risky action R. The safe action always guarantees a payoff of zero. The risky action yields a payoff $\alpha > 0$ only if the state is H and all his neighbors coordinate on the risky action; otherwise, it incurs a loss of $\beta > 0$. Formally, the payoff of an agent i under state θ and action profile $a \in \{S, R\}^N$ is given by

$$u_i(a, \theta) = \begin{cases} 0, & \text{if } a_i = S, \\ \alpha, & \text{if } a_i = R, \theta = H \text{ and } a_j = R \ \forall j \in N_i, \\ -\beta, & \text{otherwise.} \end{cases}$$

The agents are uncertain about the state and share a common prior. For simplicity, we assume $\mathbb{P}(\theta = H) = \mathbb{P}(\theta = L) = 1/2$. Each agent i receives a signal s_i about the state. Conditional on the state, the signals are independent and identically distributed. We assume that the signals take value in a finite set O, and denote by μ_H, μ_L the conditional distributions of the signals:

$$\mu_H(o) = \mathbb{P}(s_i = o | \theta = H) \qquad \mu_L(o) = \mathbb{P}(s_i = o | \theta = L).$$

We assume that both μ_H and μ_L have full support, equal to O. That is, a signal can never reveal the state completely. We also assume that $\mu_H \neq \mu_L$, i.e., that signals are informative.

The state and signals are random variables in the probability space $(\Omega, \mathcal{F}, \mathbb{P})$, where $\Omega = \{H, L\} \times O^N$, $\mathcal{F}$ is the product sigma-algebra, and $\mathbb{P}$ is the measure described above.

Before choosing an action, agents participate in a communication phase. In this phase, each agent i observes the signals of all agents in $B_r(i)$, for some fixed radius of observation r. Communication is non-strategic, with agents simply revealing their signals. An alternative interpretation is that signals are not private, but rather public and local, so that each can be directly observed without the need for communication, but only by nearby agents.

We denote the set of possible observations of agent i with radius of observation r by $O_i^r = O^{B_r(i)}$. The information available to agent i is the random variable $I_i^r = (s_j)_{j \in B_r(i)}$, which takes values in O_i^r. Thus, a pure strategy for agent i is a map from O_i^r to $\{S, R\}$. This defines a Bayesian game, and accordingly our solution concept is that of Bayes-Nash equilibrium.

Since the network is infinite and connected, the size of $B_r(i)$ goes to infinity as r increases, and so the number of signals observed by each agent in the communication phase increases. Since signals are informative about the state, agents individually learn the state. In particular, as r tends to infinity, each agent's posterior belief $\mathbb{P}(\theta = H | I_i^r)$ converges to 1 if $\theta = H$, and converges to 0 if $\theta = L$, i.e. for all $\varepsilon > 0$,

$$\lim_{r \to \infty} \mathbb{P}\Big(\mathbb{P}(\theta = H | I_i^r) > 1 - \varepsilon \Big| \theta = H\Big) = \lim_{r \to \infty} \mathbb{P}\Big(\mathbb{P}(\theta = H | I_i^r) < \varepsilon \Big| \theta = L\Big) = 1.$$

We focus on pure Bayes-Nash equilibria, which are characterized by a collection of sets $(X_i^r)_{i \in N}$, where $X_i^r \subseteq O_i^r$. These are the sets on which i chooses the risky action, i.e., $a_i = R$ if and only if $I_i^r \in X_i^r$. The expected utility of agent i from choosing the risky action upon observing $I_i^r = x$ is

$$\alpha \mathbb{P}(\theta = H, I_j^r \in X_j^r \ \forall j \in N_i | I_i^r = x) - \beta(1 - \mathbb{P}(\theta = H, I_j^r \in X_j^r \ \forall j \in N_i | I_i^r = x)),$$

whereas choosing a safe action yields zero.

Therefore, the strategy profile induced by $(X_i^r)_{i \in N}$ is an equilibrium if and only if, for every $i \in N$,

$$\forall x \in X_i^r, \quad \mathbb{P}\left(\theta = H, I_j^r \in X_j^r \ \forall j \in N_i | I_i^r = x\right) \geq \frac{\beta}{\alpha + \beta},$$
$$\forall y \notin X_i^r, \quad \mathbb{P}\left(\theta = H, I_j^r \in X_j^r \ \forall j \in N_i | I_i^r = y\right) \leq \frac{\beta}{\alpha + \beta}.$$

Denoting $q = \beta/(\alpha + \beta)$, the risky action is optimal whenever the probability that the state is high and all neighbors choose the risky action is at least q. We resolve ties in favor of R, assuming that agents take the risky action when they are indifferent, so that the second inequality above will always be strict.

We ask: on which networks can agents coordinate efficiently in equilibrium when they observe signals within a sufficiently large radius r? That is, we want to know

whether it holds that with high probability, they choose the risky action in the high state and the safe action in the low state.

2.1 Network q-belief and Network Common Learning

For any event $E \subseteq \Omega$, define

$$\mathcal{B}_{i,r}^q(E) = \{\omega \in \Omega : \mathbb{P}(E|I_i^r)(\omega) \geq q\}.$$

This is the event on which agent i, using signals in $B_r(i)$, assigns probability at least q to E.[1] If each neighbor j of i chooses the risky action on X_j^r, then $\mathcal{B}_{i,r}^q(H \cap (\bigcap_{j \in N_i} X_j^r))$ is the event on which agent i assigns a probability at least q to the event that the state is high and every neighbor of i chooses risky action.[2] Therefore, the equilibrium condition can equivalently be written as

$$X_i^r = \mathcal{B}_{i,r}^q \left(H \cap \left(\bigcap_{j \in N_i} X_j^r \right) \right) \tag{1}$$

for every agent i and radius of observation r.

This is a fixed-point condition. To identify its solutions, we study iterated network beliefs. These are defined recursively for $n \geq 0$. For $n = 0$, let $C_{i,r}^{q,0} = \Omega$, the entire sample space. For $n \geq 1$, define

$$C_{i,r}^{q,n}(H) = \mathcal{B}_{i,r}^q \left(H \cap \left(\bigcap_{j \in N_i} C_{j,r}^{q,n-1}(H) \right) \right).$$

That is, starting from the entire sample space Ω, at each subsequent step the event is defined to be the event that the agent q-believes that the state is high and the previous-level condition holds for every neighbor.

The event $C_{i,r}^{q,n}(H)$ is the event that agent i has n^{th} order network q-belief. In particular,

$$C_{i,r}^{q,1}(H) = \mathcal{B}_{i,r}^q(H)$$

is the event in which agent i assigns probability at least q to the high state. Likewise,

$$C_{i,r}^{q,2}(H) = \mathcal{B}_{i,r}^q \left(H \cap \left(\bigcap_{j \in N_i} \mathcal{B}_{j,r}^q(H) \right) \right)$$

is the event that agent i has second order network q-belief: i assigns probability at least q to the joint event that state is H and all his neighbors q-believe that the state

[1] Since $B_r(i)$ is finite, $\sigma(I_i^r)$ is generated by the finite partition $\{\{I_i^r = x\} : x \in O_i^r\}$, in which every atom has positive probability. Hence the conditional probability $\mathbb{P}(E|I_i^r)$ admits a canonical version, given on each atom by $\mathbb{P}(E \cap \{I_i^r = x\})/\mathbb{P}(I_i^r = x)$, and $\mathcal{B}_{i,r}^q(E)$ is well defined as the union of the atoms on which this value is at least q.

[2] Here and below we use X_j^r to denote the event $\{I_j^r \in X_j^r\}$ and H to denote the event $\{\theta = H\}$.

is H. At higher levels, the same condition is applied recursively to i's neighbors, their neighbors and so on.

The event that i has network q-belief is

$$C_{i,r}^q(H) = \bigcap_{n \geq 0} C_{i,r}^{q,n}(H).$$

This is the event on which q-belief in H persists through every finite order of beliefs, starting from i. The following lemma shows that the limiting event satisfies the same fixed-point condition required for equilibrium. To simplify notation, we henceforth omit the (H) and write $C_{i,r}^{q,n}$ and $C_{i,r}^q$ to indicate $C_{i,r}^{q,n}(H)$ and $C_{i,r}^q(H)$.

Lemma 1. *For every i and r,*

$$C_{i,r}^q = \mathcal{B}_{i,r}^q \left(H \cap \left(\bigcap_{j \in N_i} C_{j,r}^q \right) \right).$$

Thus, network q-belief is *self-sustaining*: whenever it holds at i, agent i assigns probability at least q to the joint event that the state is H and network q-belief holds at each of i's neighbors.

Since the family of sets $(C_{i,r}^q)_{i \in N}$ satisfies the equilibrium condition (1), it induces an equilibrium. The next proposition shows that this equilibrium has the largest possible risky set: whenever another equilibrium prescribes the risky action to agent i, this equilibrium does so as well. Recall that we describe agent i's strategy by the set X_i^r on which he chooses the risky action.

Proposition 1. *The family $(C_{i,r}^q)_{i \in N}$ induces an equilibrium in which $X_i^r = C_{i,r}^q$. Moreover, it is the maximal risky equilibrium in the sense that in any other equilibrium, $X_i^r \subseteq C_{i,r}^q$.*

We now study whether network q-belief in H emerges as agents observe larger neighborhoods. Particularly, this requires that the events $(C_{i,r}^q)_{i \in N}$ become very likely for every agent in state H. This motivates the following notion of network common learning.

Definition 1. *Network q-learning of H occurs if for every agent i,*

$$\lim_{r \to \infty} \mathbb{P}\left(C_{i,r}^q \middle| \theta = H \right) = 1.$$

Network common learning of H occurs if network q-learning of H occurs for all $q \in (0,1)$.

So, network q-learning means that in state H network q-belief in H occurs with probability approaching 1 for every agent. It also implies asymptotic efficiency of the maximal risky equilibrium. Since the maximal equilibrium prescribes the risky action on these events, risky play becomes asymptotically likely in state H. Individual learning implies that in state L each agent chooses the safe action with high probability. For choosing the safe action, no higher-order beliefs are necessary: once the first-order belief in H is below q, agents will choose the safe action.

3 Coordination Failure on Networks with Bottlenecks

In this section we study graphs on which the only equilibrium is the one in which agents always play the safe action, for any radius of observation. We show that a geometric condition on the graph is sufficient to guarantee this: the graph must contain a long path along which neighboring agents' observation neighborhoods differ by only a uniformly bounded number of signals. On such graphs, network common learning is not attained.

Definition 2. *Fix $r, m > 0$. A path of distinct agents $i_1, i_2, \ldots, i_T$ in the graph G is an (r, m)-bottleneck if*

1. *$|B_r(i_{t+1}) \backslash B_r(i_t)| \leq m$ for all $t < T$, i.e., agent i_{t+1} has no more than m agents in their r-neighborhood that are not in agent i_t's r-neighborhood.*

2. *$B_r(i_1) \cap B_r(i_T) = \emptyset$, i.e., the first and last agent's r-neighborhoods are disjoint.*

The first condition in the definition is that along the path, neighboring agents' r-neighborhoods are highly overlapping: moving from i_t to i_{t+1} introduces at most m new signals. The second condition ensures that the path is long enough so that all signals observed by the first agent are not observed by the last; equivalently, that i_1 and i_T are more than distance $2r$ apart. A simple example is the line graph, which contains an $(r, 1)$-bottleneck for all r.

Our main result in this section is that on graphs with bottlenecks coordination is impossible. Denote by $p_0 = \min_{o \in O} \mu_H(o)$ the probability of the least likely signal, under the high state.

Theorem 1. *Suppose that G contains an (r, m)-bottleneck. If q is large enough (it suffices that $q > 1 - p_0^m$), then $C_{i,r}^q = \emptyset$ for all i: network q-belief does not occur and in the unique equilibrium all agents choose the safe action.*

As mentioned above, the line graph has a uniform family of bottlenecks, i.e., an $(r, 1)$-bottleneck for all r. An immediate corollary of Theorem 1 is that the existence of such a uniform family excludes network common learning.

Corollary 1. *Suppose that there exists an $m > 0$ such that for every $r > 0$ the graph contains an (r, m)-bottleneck. Then network common learning is not obtained.*

The proof of Theorem 1 appears in Appendix A.3. The idea behind it is the following argument, which is similar to infection arguments used in the literature (e.g., Carlsson and van Damme, 1993).

For this proof sketch, consider the one-sided line graph, i.e., $N = \mathbb{N}$ and $i, j \in N$ are neighbors if $|i - j| = 1$. Fix an $r > 0$ and an agent i, and let $x_i \in O_i^r$ be an arbitrary observation by agent i. To prove the claim, we show that after observing x_i, agent i does not have network q-belief, and therefore must take the safe action.

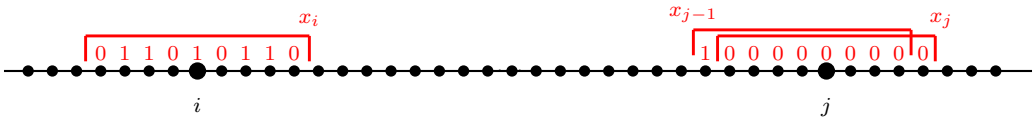


Figure 1: Proof sketch of Theorem 1. Zeros indicate signal realizations that induce low beliefs, while ones induce high beliefs. The observation x_i is arbitrary, while x_j is fixed to induce a low belief. Conditioned on $j-1$ observing x_{j-1}, there is sufficiently high probability that j observed x_j, ruling out network q-belief by $j-1$. By induction, this rules out network q-belief by i, for arbitrary x_i.

To this end, consider some agent j who is far enough from agent i so that their observation neighborhoods are disjoint: $B_r(i) \cap B_r(j) = \emptyset$ ($j = i + 2r + 1$ suffices). Let $x_j \in O_j^r$ be any observation by j which implies that j does not have network q-belief. For example, this holds for any x_j which induces a posterior belief that is lower than q: $\mathbb{P}(\theta = H | I_j^r = x_j) < q$. Fix any signal realizations for the agents $i+1, i+2, \ldots, j-1$ that are compatible with x_i and x_j, and denote the observations by these agents by $x_{i+1}, \ldots, x_{j-1}$; see Figure 1.

The key observation is that because x_j does not induce network q-belief for agent j, from agent $j-1$'s point of view, this $(I_j^r = x_j)$ happens with probability at least p_0: only one signal observed by j is not observed by $j-1$, and so the probability that $I_j^r = x_j$ conditioned on $I_{j-1}^r = x_{j-1}$ is bounded away from zero, regardless of how large r is. It follows that x_{j-1} does not induce network q-belief in $j-1$. By induction, x_i does not induce network q-belief in agent i. Since i and x_i are arbitrary, this completes the proof.

4 Efficient Coordination on Networks with Large Neighborhood Growth

In this section we identify two classes of graphs on which sufficiently fast uniform growth of observation neighborhoods guarantees network common learning. The first result applies to "convex" subgraphs of grids and the second to bounded-degree trees. In the end of the section we describe the common mechanism behind them.

4.1 Convex Subgraphs of Grids

We first consider subgraph of grids that satisfy a convexity condition. In dimension k, the grid is the graph with vertex set $\mathbb{Z}^k$ and edges between $i, j \in N$ if $|i - j| = 1$. In particular, each agent has $2k$ neighbors. We say that a graph $G = (N, E)$ is an induced subgraph of the k-dimensional grid if the vertex set N is a subset of $\mathbb{Z}^k$ and the edges are inherited from $\mathbb{Z}^k$, i.e., $i, j \in N$ are neighbors if $|i - j| = 1$.

Definition 3. *A graph $G = (N, E)$ is a* convex subgraph of a grid *if it is an induced*

subgraph of the k-dimensional grid and there is a convex subset $W \subseteq \mathbb{R}^k$ *such that* $N = \mathbb{Z}^k \cap W$.

Convexity implies that graph distances in G coincide with graph distances in the larger graph $\mathbb{Z}^k$ (this is a weaker condition, which in fact is sufficient for our results). The class includes $\mathbb{Z}^k$ itself, as well as many subgraphs of the grid, including half-spaces, orthants and convex horns.

Another example of a convex subgraph of a grid is of course $\mathbb{Z}$ itself. Thus, by Theorem 1, convexity is not sufficient for network learning. We will need an additional condition, namely that for large r, the ball of radius r around an agent i is large: it needs to contain much more than $r \log r$ agents.

Formally, we define the *neighborhood growth function* of a graph by

$$\mathcal{G}(r) = \inf_{i \in N} |B_r(i)|,$$

so $\mathcal{G}(r)$ is the size of the smallest ball of radius r. We say that graph grows faster than some function $f : \mathbb{N} \to \mathbb{N}$ if $\lim_r \mathcal{G}(r)/f(r) = \infty$. For example, for the two-dimensional grid $\mathcal{G}(r)$ is roughly $2r^2$, and so $\mathbb{Z}^2$ grows faster than $r \log r$. For the line graph $\mathcal{G}(r) = 2r + 1$, and so it does not grow faster than $r \log r$.

Theorem 2. *Suppose that* G *is a convex subgraph of a grid and that it grows faster than* $r \log r$. *Then network common learning is obtained on* G.

Thus, within the class of convex subgraphs of grids, network common learning follows whenever the sizes of observation neighborhoods grow uniformly faster than $r \log r$. In particular this includes $\mathbb{Z}^k$ for $k \geq 2$, as well as half-spaces and orthants. Another example is the subgraph of $\mathbb{Z}^2$ with vertex set $N = \{(x, y) \in \mathbb{Z}^2 : 0 \leq x, 0 \leq y \leq \sqrt{x}\}$.

4.2 Trees

Our second result applies to trees: connected graphs that contain no cycles. We will be interested in bounded degree trees: A graph has bounded degree if there is some $d \in \mathbb{N}$ such that $|N_i| \leq d$ for all $i \in N$. An example of a bounded degree tree is the d-regular tree, or the unique connected graph with no cycles in which each vertex has d neighbors. For $d = 2$ this is isomorphic to $\mathbb{Z}$. The graph $\mathbb{Z}^2$ is clearly not a tree.

Although their geometry is different from that of subgraphs of grids, network common learning is also obtained on fast growing trees.

Theorem 3. *Suppose that* G *is a bounded-degree tree and grows faster than* r^2. *Then network common learning is obtained on* G.

Compared with the previous case, trees require faster neighborhood growth: the size of $B_r(i)$ must dominate r^2. In particular, this condition holds the infinite binary tree, and for all d-regular trees with $d \geq 3$.

A particularly nice class of trees are trees without leaves. A *leaf* of a tree is a vertex that has only one neighbor. Regular trees have no leafs, but of course the class of trees without leafs is much larger, and is in fact uncountable. For this class we obtain a sharp characterization, as a corollary of Theorem 3:

Corollary 2. *Suppose that G is a bounded degree tree without leaves. Then network common learning is obtained on G if and only if G has no $(r,1)$-bottlenecks for all r large enough.*

The only-if part follows from Corollary 1. For converse, the absence of $(r,1)$-bottlenecks implies that there is M such that every path of length M contains a vertex of degree at least 3. Thus, the number of descendants at least doubles every M levels, i.e. $|B_r(i)| \geq c2^{\lfloor r/M \rfloor}$ uniformly in i for some $c > 0$. Thus, the tree has exponential growth and the result follows from Theorem 3.

To describe the main ideas behind both Theorem 2 and Theorem 3, we sketch a proof for the case of $\mathbb{Z}^2$. We construct a family of self-sustaining events that occur with high probability in state H; a similar construction (with some additional complications) is used in the Appendix to prove the two theorems.

To define these self-sustaining events, we introduce *test regions*. Given a finite set $W \subset N$ of agents, denote by $Q_W = \cap_{j \in W} B_r(j)$ the intersection of their observation neighborhoods (see Figure 2). The signals in Q_W precisely form the information that is common knowledge among the agents in W. The test regions will be sets of the form Q_W, where W is a connected subset of the network, i.e., for any $i, j \in W$ there is a path between i and j that includes only members of W. The set of test regions is

$$\mathcal{Q}^r = \{Q_W \,:\, W \text{ is finite and connected}\}.$$

For agent i, the relevant sub-collection is

$$\mathcal{Q}_i^r = \{Q_W \in \mathcal{Q}^r \text{ such that } i \in W\}.$$

On the line graph, test regions are simply the finite intervals of length at most $2r+1$. On the two-dimensional grid, where balls are diamond-shaped, these are 45-degree slanted rectangles (see Figure 2 again).

Conditioned on the signals in a test region Q, the probability of H is determined by the sum of the log-likelihood ratios. Indeed, given a test region Q, let

$$S(Q) = \sum_{j \in Q} \log \frac{\mu_H(s_j)}{\mu_L(s_j)}$$

be the log-likelihood ratio induced by the signals in Q. The expectation of each signal's log-likelihood ratio in the high state is

$$\kappa = \sum_{o \in O} \mu_H(o) \log \frac{\mu_H(o)}{\mu_L(o)},$$

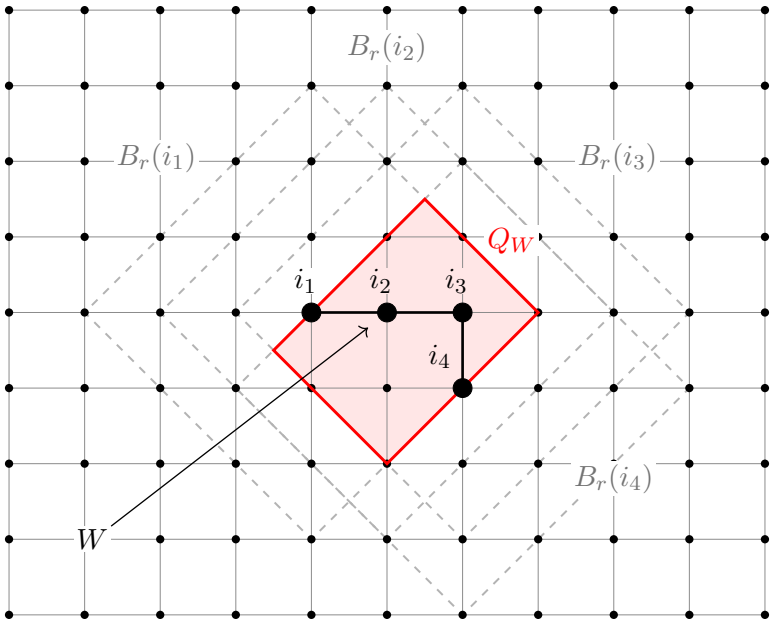


Figure 2: The test region Q_W in $\mathbb{Z}^2$ for $W = \{i_1, i_2, i_3, i_4\}$ is shown in red. It is the intersection of the agents' observation neighborhoods $Q_W = B_r(i_1) \cap B_r(i_2) \cap B_r(i_3) \cap B_r(i_4)$, shown by the dashed lines.

and so $S(Q)$ has expectation $\kappa|Q|$, conditional on the high state. For large Q, concentration of measure makes substantially smaller values unlikely. Since signals are informative, $\kappa > 0$.

The sets $(\mathcal{Q}_i^r)_i$ form the minimal collections such that $B_r(i) \in \mathcal{Q}_i^r$ and, for any j who is a neighbor of i and $Q \in \mathcal{Q}_j^r$ it holds that $B_r(i) \cap Q \in \mathcal{Q}_i^r$. The importance of test regions in this setting stems from this property, and their close connection to the agents' higher order beliefs. The first order belief of agent i in the event H is determined by $S(B_r(i))$, and $B_r(i)$ is indeed one of i's test regions. If j is i's neighbor, then i's belief in j's belief in H is determined by $S(B_r(i))$ and $S(B_r(i) \cap B_r(j))$—another test region. More generally, if Q is a test region of j, then i's belief about what j observes in Q is entirely determined by $S(B_r(i))$ and $S(B_r(i) \cap Q)$.

Using test regions, we can now define our self-sustaining events. Choose $\varepsilon > 0$ small enough. The events are

$$Y_i^r = \left\{ S(Q) \geq \varepsilon\kappa|Q| - \varepsilon^2 r^2 \quad \forall Q \in \mathcal{Q}_i^r \right\}.$$

That is, Y_i^r is the event that in each test region Q the log-likelihood of the signals exceeds some threshold. Recall that the expectation of $S(Q)$ is $\kappa|Q|$. The threshold is lower than this expectation by both the ε multiplicative factor, and by the $\varepsilon^2 r^2$ additive value. The multiplicative factor means that when Q is large, concentration of measure implies that (in the high state) $S(Q)$ will very likely be higher than this threshold. For small enough sets the additive factor makes the threshold so low that

it is satisfied automatically, with probability one. Thus, in the high state, each event $\{S(Q) \geq \varepsilon\kappa|Q| - \varepsilon^2 r^2\}$ will occur with probability one or close to it.

More important, however, is that on Y_i^r agent i assigns high probability to Y_j^r for any neighbor j. That is, these sets are self-sustaining. Indeed, let i and j be neighbors, and let Q be a test region of agent j. On Y_i^r, we show that $\{S(Q) \geq \varepsilon\kappa|Q| - \varepsilon^2 r^2\}$ with high probability. The idea is depicted in Figure 3. Let $T = Q \cap B_r(i)$ be the part of Q also observed by i, and let $F = Q \setminus B_r(i)$ be the (j,i)-*information boundary* of Q: the part that is not observed by i. Since i is a neighbor of j, then T is also a test region of agent i. Thus, on Y_i^r the signals on T are strong, and the remaining uncertainty comes from F.

The additive factor $\varepsilon^2 r^2$ means that when Q is small (say, of order less than r^2, which is roughly the size of balls of radius r) then $S(Q)$ is automatically above the required threshold. When Q is of order r^2, the information boundary F is of order r. In particular, when r is large, the information boundary is large. Then concentration of measure kicks in, and player i deems it very unlikely that the signals in F are very negative, tipping $S(Q)$ below the required threshold.

The proof fails in the case of a graph with bottlenecks like $\mathbb{Z}$, since there the boundary F is always of bounded size, and so the benefit of concentration of measure is not attained even for large r. This idea does work for convex subgraphs of grids and for trees, although a more delicate specification of the thresholds is required.

5 General Sufficient Conditions for Network Common Learning

In this section we go beyond the specific families of convex subgraphs of grids and tree, and give general conditions that suffice for network common learning. These conditions formalized the same intuitions used in the previous section, but apply to a much larger class of graphs.

Suppose i and j are neighbors in the graph. Recall from the previous section that given a test region $Q \in \mathcal{Q}_i^r$ (observed by i), we call $Q \setminus B_r(j)$ the (i,j)-*information boundary* of Q (see Figure 3). This is the part of Q that is not observed by j. Our condition of *large information boundaries* requires that when Q is large, its information boundaries are large, or else the boundaries are empty.

Definition 4. *The social network graph has* (ε, δ)-large information boundaries *if for all $r > 0$ and neighbors i, j it holds that for every test region $Q \in \mathcal{Q}_i^r$ of size at least $\varepsilon|B_r(i)|$, the (i,j)-information boundary $Q \setminus B_r(j)$ is either empty, or else is of size at least $\delta \cdot r$.*

The social network graph has large information boundaries *if for all $\varepsilon > 0$ there is a $\delta > 0$ such that the graph has (ε, δ)-large information boundaries.*

If the boundary is empty, higher-order beliefs are trivial: Q is completely observed

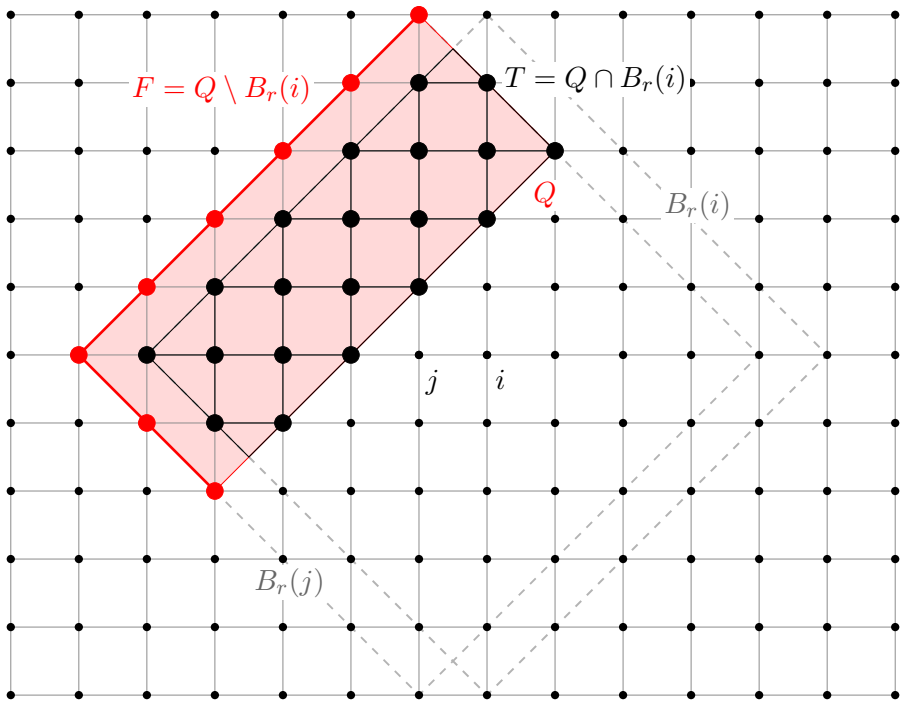


Figure 3: A test region $Q \subset \mathbb{Z}^2$ belonging to agent j is shown as the shaded region. The observation neighborhoods $B_r(i)$ and $B_r(j)$ are shown by dashed lines. The highlighted black agents form $T = Q \cap B_r(i)$, the part of Q observed by both i and j. The red agents form the (j,i)-information boundary $F = Q \setminus B_r(i)$, the part of Q observed by j but not by i.

by j. Otherwise, this property requires that the unobserved part is large, which will help us control j's higher-order beliefs by concentration measure.

For network common learning we will need two additional geometric conditions. The first is that the collection of test regions $\mathcal{Q}_i^r$ is not too large.

Definition 5. *The social network graph has* subexponential-$\mathcal{Q}$ *if the size of the collection* $(\mathcal{Q}_i^r)_{i,r}$ *satisfies* $\lim_{r\to\infty} \left(\sup_{i\in N} |\mathcal{Q}_i^r|\right)^{1/r} = 1$.

The second is that the size of neighborhoods does not vary too much between agents.

Definition 6. *The social network graph has* similar neighborhood sizes *if there is a constant* $K \geq 1$ *such that for all* $r > 0$ *and for all* $i, j \in N$ *it holds that* $|B_r(i)|/|B_r(j)| \leq K$.

It is straightforward to verify that all three of these conditions are satisfied by the two-dimensional grid,[3] and, more generally, by higher dimensional grids. These

[3]In the coordinates $u = x+y$, $v = x-y$, the ball $B_r(i)$ of $\mathbb{Z}^2$ is a box $\{|u-u_i| \leq r\} \cap \{|v-v_i| \leq r\}$,

graphs are already covered by the previous theorem, but this condition applies more generally to many graphs that are much less rigidly structured.

Theorem 4. *Suppose that G satisfies the large information boundary condition, the subexponential-$\mathcal{Q}$ condition, and the similar neighborhood sizes condition. Then network common learning is obtained on G.*

The theorem follows from the construction described in Section 4, letting the threshold be $\varepsilon\kappa|Q| - \varepsilon \sup_{j\in N} |B_r(j)|$. The subexponential-$\mathcal{Q}$ condition ensures that there are sufficiently few test regions for the required concentration bounds to hold simultaneously, while similar neighborhood sizes condition ensures that the slack remains small relative to each agent's observation neighborhood. Together these conditions yield network common learning.

6 Finite Networks

7 Conclusion

We study global games, played locally on a social network. Agents receive a large amount of information about the state by learning the private signals of agents in their r-neighborhood. As in the classic literature, we ask whether higher-order beliefs become aligned. In particular, we study network common learning, the network analogue of common learning. The answer in our case is that it depends non-trivially on the network geometry.

We do not provide a full characterization of graphs on which network common learning is obtained, instead providing separate necessary and sufficient conditions. A major point for future research is to close this gap. A natural question is whether network common learning is a *quasi-isometry* invariant. That is, is network common learning robust to local changes in the graph that do not change the global geometry?[4]

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so test regions are boxes $[u_1, u_2] \times [v_1, v_2]$. Similar neighborhood sizes holds with $K = 1$ since the graph looks the same everywhere. A test region contained in $B_r(i)$ is determined by four coordinates in a window of length $2r + 1$, so $|\mathcal{Q}_i^r| \leq (2r + 1)^4$, giving subexponential-$\mathcal{Q}$. For large boundaries, note that $|B_r(i)| \geq 2r^2$, so a box Q with $|Q| \geq \varepsilon|B_r(i)|$ and side lengths at most $2r + 1$ has both sides of length at least $\varepsilon r/2$. Moving to a neighbor j shifts the box $B_r(j)$ by one unit in u and in v; hence if $Q \setminus B_r(j)$ is nonempty, Q protrudes past a face of $B_r(j)$, and the boundary contains a full side of Q, of size at least δr with $\delta = \varepsilon/4$.

[4]Two connected graphs $G_1 = (N_1, E_1)$ and $G_2 = (N_2, E_2)$ are isometric (or isomorphic) if there is a bijection $\varphi\colon N_1 \to N_2$ such that $d_1(i, j) = d_2(\varphi(i), \varphi(j))$, where d_i denote graph distances. They are quasi-isometric if there is a map $\varphi\colon N_1 \to N_2$ and constants $A \geq 1$, $B \geq 0$ such that for every $i, j \in N_1$

$$\frac{1}{A} d_1(i, j) - B \leq d_2(\varphi(i), \varphi(j)) \leq A d_1(i, j) + B,$$

and every element of N_2 is of distance at most B from the image of φ.

Note that all of the assumptions of Theorem 4 are about the geometry of the network, rather than the structure of signals; we only assume that signals are finitely supported, informative but not revealing. The same holds for Theorem 1, and even more saliently for its Corollary 1. This suggests a natural follow-up question, namely whether network common learning is a purely geometric property, or whether it can also depend on the signals. That is, on a graph where network common learning is obtained for some (finitely supported, informative and non-revealing) signal distribution, is it also obtained for all other such distributions?

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A Proofs and Auxiliary Results

A.1 Proof of Lemma 1

We first claim that the sequence of sets $(C^{q,n}_{j,r})_n$ is decreasing. This follows from the fact that the $\mathcal{B}^q_{i,r}$ operator is monotone with respect to set inclusion.

Let $E_n = H \cap \left(\bigcap_{j\in N_i} C^{q,n}_{j,r}\right)$. Since $(C^{q,n}_{j,r})_n$ is decreasing, $(E_n)_{n\geq 0}$ is also decreasing. Hence, $E_n \downarrow \cap_n E_n$. Next, by continuity from above,

$$\mathbb{P}(\cap_{n\geq 0} E_n | I^r_i) = \lim_{n\to\infty} \mathbb{P}(E_n|I^r_i) = \inf_{n\geq 0} \mathbb{P}(E_n|I^r_i).$$

Thus,

$$\omega \in \bigcap_{n\geq 0} \mathcal{B}^q_{i,r}(E_n) \iff \forall n:\ \mathbb{P}(E_n|I^r_i)(\omega) \geq q \iff \inf_{n\geq 0} \mathbb{P}(E_n|I^r_i)(\omega) \geq q \iff$$
$$\iff \mathbb{P}(\cap_{n\geq 0} E_n|I^r_i)(\omega) \geq q \iff \omega \in \mathcal{B}^q_{i,r}(\cap_{n\geq 0} E_n).$$

So, $\bigcap_{n\geq 0} \mathcal{B}^q_{i,r}(E_n) = \mathcal{B}^q_{i,r}(\cap_{n\geq 0} E_n)$. Finally,

$$C^q_{i,r} = \Omega \cap \bigcap_{n\geq 1} C^{q,n}_{i,r} = \bigcap_{n\geq 0} C^{q,n+1}_{i,r} = \bigcap_{n\geq 0} \mathcal{B}^q_{i,r}(E_n) = \mathcal{B}^q_{i,r}(\cap_{n\geq 0} E_n) =$$
$$= \mathcal{B}^q_{i,r}\left(\bigcap_{n\geq 0}\left(H \cap \bigcap_{j\in N_i} C^n_{j,r}\right)\right) = \mathcal{B}^q_{i,r}\left(H \cap \bigcap_{j\in N_i}\bigcap_{n\geq 0} C^n_{j,r}\right) = \mathcal{B}^q_{i,r}\left(H \cap \bigcap_{j\in N_i} C_{j,r}\right),$$

which completes the proof.

A.2 Proof of Proposition 1

Firstly, note that by Lemma 1 the family $(C^q_{i,r})_{i\in N}$ satisfies the equilibrium condition (1). Thus, it induces an equilibrium.

Next, we prove by mathematical induction that for every agent i, $X^r_i \subseteq C^{q,n}_{i,r}$ for every n and thus $X^r_i \subseteq \bigcap_{n\geq 0} C^{q,n}_{i,r} = C^q_{i,r}$.

For $n = 0$, $X^r_i \subseteq \Omega = C^{q,0}_{i,r}$. Now suppose that for every agent j, $X^r_j \subseteq C^k_{j,r}$ for $k \leq (n-1)$. Then,

$$H \cap \left(\bigcap_{j\in N_i} X^r_j\right) \subseteq H \cap \left(\bigcap_{j\in N_i} C^{n-1}_{j,r}\right).$$

Since the belief operator is monotone,

$$X^r_i = \mathcal{B}^q_{i,r}\left(H \cap \left(\bigcap_{j\in N_i} X^r_j\right)\right) \subseteq \mathcal{B}^q_{i,r}\left(H \cap \left(\bigcap_{j\in N_i} C^{q,n-1}_{j,r}\right)\right) = C^{q,n}_{i,r}.$$

Finally, consider any mixed strategy $\sigma_i : I^r_i \to [0,1]$ with $\sigma_i(x)$ denoting the probability of choosing action R on observing x and denote $Y = \{x \in O^r_i : \sigma(x) > 0\}$ as the observation on which action R is played with positive probability, we know $Y \subseteq \mathcal{B}^q_{i,r}\left(H \cap \left(\bigcap_{j\in N_i} X^r_j\right)\right)$ and the proof for the case of mixed strategy is essentially the same as above.

A.3 Proof of Theorem 1

Consider two agents i, j, and signal realizations $x_i \in O_i^r$ and $x_j \in O_j^r$ for each agent. We say that x_i and x_j are *compatible* if they agree on the overlapping region $B_r(i) \cap B_r(j)$.

By the definition of $C_{i,r}^{q,n}$, the event that i has n^{th} order network q-belief, $C_{i,r}^{q,n}$ is $\sigma(I_i^r)$-measurable. That is, this event depends only on the information available to agent i. As such, we can identify this event with a subset of O_i^r, the possible observations of agent i. Accordingly, for $x \in O_r^i$ we write $x \in C_{i,r}^{q,n}$ whenever agent i has n^{th} order network q-belief upon observing x.

By the theorem hypothesis, there is an (r, m)-bottleneck path $i_1, i_2, \ldots, i_T$. Consider two adjacent agents along the bottleneck path, i_t and i_{t+1}. Let $x_t \in O_{i_t}^r$ and $x_{t+1} \in O_{i_{t+1}}^r$ be compatible. Suppose that $x_{t+1} \notin C_{i_{t+1},r}^q$, i.e., when agent i_{t+1} observes x_{t+1}, i_{t+1} does not have network q-belief. We claim that this implies that $x_t \notin C_{i_t}^q$, i.e., agent i_t does not have network q-belief upon observing the compatible x_t. Indeed, since $|B_r(i_{t+1}) \setminus B_r(i_t)| \leq m$, and the signals observed by i_{t+1} but not by i_t are independent conditional on the state, it follows that

$$\mathbb{P}(I_{i_{t+1}}^r = x_{t+1} | I_{i_t}^r = x_t, H) = \mathbb{P}(\{s_l = x_{i_{t+1},l} \; \forall l \in (B_r(i_{t+1}) \setminus B_r(i_t))\} | H) \geq p_0^m,$$

where $p_0 = \min_{o \in O} \mu_H(o)$.

By our assumption that $x_{t+1} \notin C_{i_{t+1}}^q$,

$$\mathbb{P}\left(C_{i_{t+1},r}^q \middle| I_{i_t}^r = x_t, H\right) \leq 1 - \mathbb{P}(I_{i_{t+1}}^r = x_{t+1} | I_{i_t}^r = x_t, H) \leq 1 - p_0^m,$$

and hence

$$\begin{aligned}\mathbb{P}\left(H \cap \bigcap_{j \in N_{i_t}} C_{j,r}^q \middle| I_{i_t}^r = x_t\right) &\leq \mathbb{P}\left(C_{i_{t+1},r}^q \middle| I_{i_t}^r = x_t, H\right) \mathbb{P}(H | I_{i_t}^r = x_t) \\ &\leq 1 - p_0^m < q.\end{aligned}$$

Thus

$$x_t \notin \mathcal{B}_{i_t,r}^q \left(H \cap \left(\bigcap_{j \in N_{i_t}} C_{j,r}^q\right)\right) = C_{i_t,r}^q,$$

by the fixed-point property of $C_{i,r}^q$ (Lemma 1). We have thus shown that

$$x_{t+1} \notin C_{i_{t+1},r}^q \implies x_t \notin C_{i_t,r}^q. \tag{2}$$

We now note that the theorem assumptions imply that $q > 1/2$. Indeed, since signals are informative, there are at least two signal realizations, and so $p_0 \leq 1/2$. Thus $q > 1 - p_0^m$ implies that $q > 1/2$. Fix an arbitrary observation $x_1 \in O_{i_1}^r$ for agent i_1. Let $x_T \in O_{i_T}^r$ be any observation that induces beliefs below q; at least one must

exist, since q is larger than the prior $1/2$. In particular, $x_T \notin C^q_{i_T,r}$, since when i_T observes x_T their posterior belief is below q, so x_T is not in $\mathcal{B}^q_{i_T,r}(H) \supseteq C^q_{i_T,r}$.

Since $B_r(i_1) \cap B_r(i_T) = \emptyset$, there exists a signal configuration on $\bigcup_t B_r(i_t)$ that agrees with x_1 on $B_r(i_1)$ and with x_T on $B_r(i_T)$. Hence, there are $x_2, \ldots, x_T$ such that $x_1, x_2, \ldots, x_T$ are all pairwise compatible.

Thus, using (2), it follows by induction that $x_1 \notin C^q_{i_1,r}$:

$$x_T \notin C^q_{i_T,r} \implies x_{T-1} \notin C^q_{i_{T-1},r} \implies \cdots \implies x_1 \notin C^q_{i_1,r}$$

Since the choice of x_1 is arbitrary, we have shown that $C^q_{i_1,r} = \emptyset$: i_1 never has network q-belief. It then follows from the fixed point property of these sets (Lemma 1), that the same holds for i_1's neighbors, and by induction for all the agents.

Finally, by Proposition 1, it follows that all agents play the safe action in every equilibrium, which completes the proof.

A.4 General Sufficient Conditions for Network Common Learning

We begin with a general theorem giving sufficient conditions for network common learning. We then verify these conditions for the three classes considered in the paper (Theorems 2, 3 and 4).

Let $l(o) = \log(\mu_H(o)/\mu_L(o))$ denote the log-likelihood ration of signal o, and let $\underline{l} = \min_{o \in O} l(o)$, $\bar{l} = \max_{o \in O} l(o)$. Recall that $\kappa = \sum_{o \in O} \mu_H(o) l(o)$ is the expected log-likelihood ratio of a signal in state H. Fix a threshold $0 < \eta < \kappa$, and let $c = \frac{2(\kappa-\eta)^2}{(\bar{\ell}-\underline{\ell})^2}$.

We define the self-sustaining events more generally here. They will take the form

$$Y^r_i = \{S(Q) \geq \eta(|Q| - \lambda_r - A_r(Q)) \quad \forall Q \in \mathcal{Q}^r_i\},$$

where λ_r is a common slack term and $A_r(Q)$ is a region specific allowance, defined below.

The idea is similar to one discussed in Section 4: every test region must contain sufficiently strong evidence for H. The difference is that we now allow the required threshold to depend on both the test region and the radius r. When the term λ_r is large enough, it makes the condition automatically satisfied for sufficiently small test regions. The allowance $A_r(Q)$, in turn, provides additional slack that is needed when comparing the test regions of neighboring agents. To see why we need it, consider a test region $Q \in \mathcal{Q}^r_j$ and j's neighbor i. Agent i observes $T = Q \cap B_r(i)$ but does not observe $F = Q \setminus B_r(i)$. If F is large, then it can be controlled by concentration. If F is small, concentration is not useful. In that case we allow for the worst possible realization, $S(F) \geq \underline{l}|F|$. Since the threshold requires roughly η units of evidence, F would ideally contribute $\eta|F|$. Thus, a small F can contribute up to $(\eta - \underline{l})|F|$ less evidence than the the benchmark. The allowance $A_r(Q)$ accumulates these adjustments along transitions between test regions.

Despite this more general form, the interpretation of Y_i^r is the same as before: every sufficiently large test region must contain sufficiently strong evidence for H. Indeed, fix any $\tilde{\eta} < \eta$, if

$$|Q| \geq \frac{\eta}{\eta - \tilde{\eta}} \left(\lambda_R + \sup_{q \in \mathcal{Q}^r} A_r(Q) \right) = \rho_r,$$

then

$$\eta(|Q| - \lambda_r - A_r(Q)) \geq \eta \left(|Q| - \lambda_r - \sup_{q \in \mathcal{Q}^r} A_r(Q) \right) \geq \tilde{\eta}|Q|.$$

Hence, for sufficiently large test regions, membership in Y_i^r implies the simpler condition $S(Q) \geq \tilde{\eta}|Q|$.

The two conditions are, however, not equivalent, and we cannot redefine Y_i^r using only this simpler threshold for the general result: the variation $A_r(Q)$ across test regions is what provides the extra slack needed when passing between test regions.

We now define the allowance formally. The first definition is used to describe the relation between test regions of neighboring agents. The second introduces the cumulative adjustment required along such relations.

Definition 7 (Transition digraph)**.** *Fix r. The transition directed graph $\mathcal{T}_r$ has nodes*

$$\{Q : Q \in \mathcal{Q}_i^r \text{ for some } i \in N\}.$$

There is a directed edge between these nodes $e = T \to Q$ if there exist neighbors i, j such that $Q \in \mathcal{Q}_j^r$, and $T = Q \bigcap B_r(i)$. For such a transition, denote $F_e = Q \setminus T$.

Given L_r, assign to each edge e the weight

$$w_r(e) = \frac{(\eta - \underline{\ell})}{\eta} |F_e| \mathbf{1}\{|F_e| \leq L_r\}.$$

Thus, only transitions for which the unobserved part F_e is too small to be controlled by concentration receive positive weight.

Definition 8 (Allowance)**.** *For every node Q of $\mathcal{T}_r$, define*

$$A_r(Q) = \sup_{\gamma \to Q} \sum_{e \in \gamma} w_r(e),$$

where the supremum is taken over directed paths in $\mathcal{T}_r$ ending at Q.

Thus, $A_r(Q)$ is the maximal cumulative adjustment required by small unobserved parts along a sequence of transitions leading to Q.

To achieve network learning we require the following three conditions to hold:

$$\inf_{i \in N} (|B_r(i)| - \lambda_r - A_r(B_r(i))) \xrightarrow[r \to \infty]{} \infty, \tag{C.1}$$

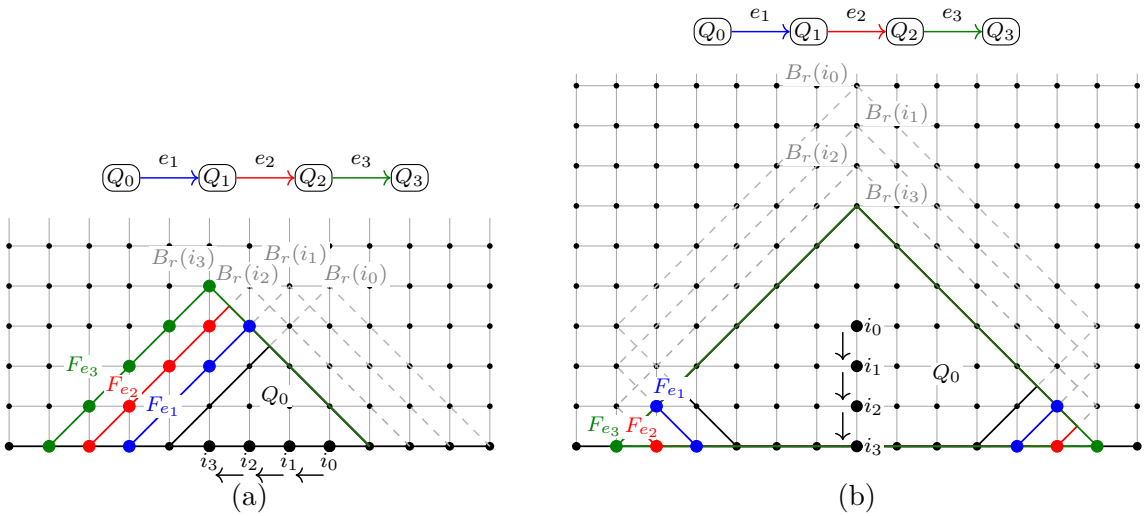


Figure 4: Examples of directed paths in the transition digraph on the half-grid. Starting from Q_0, at each step we choose a neighboring agent i_{t-1} and a region Q_t such that $Q_{t-1} = Q_t \cap B_r(i_{t-1})$. The corresponding information boundary is $F_{e_t} = Q_t \setminus Q_{t-1} = Q_t \setminus B_r(i_{t-1})$.

$$\sup_{i \in N} \sum_{j \in N_i} |\mathcal{Q}_j^r| \exp\left(-cL_r\right) \xrightarrow[r \to \infty]{} 0, \tag{C.2}$$

$$\sup_{i \in N} |\mathcal{Q}_i^r| \exp\left(-\frac{c\eta}{\eta - \underline{\ell}} \lambda_r\right) \xrightarrow[r \to \infty]{} 0. \tag{C.3}$$

Condition (C.1) requires the slack and allowance to remain small relative to the information in $B_r(i)$. Condition (C.2) ensures that the concentration on the unobserved parts larger than L_r survives the union bound over test regions. Condition (C.3) keeps the number of test regions small relative to the slack λ_r.

Theorem 5. *Suppose that conditions* (C.1), (C.2), (C.3) *are satisfied. Then network common learning is obtained.*

To prove the theorem, we first establish in three claims the properties of the events Y_i^r, and then introduce a lemma used in the final step of the argument. The first two imply that the family $(Y_i^r)_{i \in N}$ is self-sustaining, while the third shows that these events occur with probability approaching one in the high state. The lemma then links self-sustaining events to network q-belief, completing the argument for network common learning.

Claim A.1.

$$\lim_{r \to \infty} \inf_{i \in N} \inf_{x \in Y_i^r} \mathbb{P}(\theta = H \mid I_i^r = x) = 1.$$

Proof of Claim A.1. For any $i \in N$ and any $x \in Y_i^r$, since $Q = B_r(i) \in \mathcal{Q}_i^r$,

$$S(B_r(i)) \geq \eta(|B_r(i)| - \lambda_r - A_r(B_r(i))) \xrightarrow[r\to\infty]{} \infty,$$

where convergence follows from condition (C.1). Then,

$$\begin{aligned}\frac{\mathbb{P}(\theta = H | I_i^r = x)}{\mathbb{P}(\theta = L | I_i^r = x)} &= \frac{\mathbb{P}(I_i^r = x | \theta = H)}{\mathbb{P}(I_i^r = x | \theta = L)} = \prod_{j \in B_r(i)} \frac{\mu_H(s_j)}{\mu_L(s_j)} \\ &= \exp\left(\sum_{j \in B_r(i)} \log \frac{\mu_H(s_j)}{\mu_L(s_j)}\right) = \exp(S(B_r(i))) \\ &\geq \exp(\eta(|B_r(i)| - \lambda_r - A_r(B_r(i)))) \xrightarrow[r\to\infty]{} \infty.\end{aligned}$$

Hence, the posterior probability of the high state goes to 1, uniformly over i and x. □

Claim A.2.

$$\lim_{r\to\infty} \inf_{i \in N} \inf_{x \in Y_i^r} \mathbb{P}(I_j^r \in Y_j^r, \forall j \in N_i \mid I_i^r = x, \theta = H) = 1.$$

Proof of Claim A.2. Fix $i \in N$, $x \in Y_i^r$ and $j \in N_i$. To show the desired convergence, we estimate the conditional probability that $I_j^r \notin Y_j^r$. First, note that a test region Q can violate its threshold only if $|Q| > \eta\lambda_r/(\eta - \underline{\ell})$. Indeed, otherwise

$$S(Q) \geq \underline{\ell}|Q| \geq \eta(|Q| - \lambda_r) \geq \eta(|Q| - \lambda_r - A_r(Q)).$$

So, fix $Q \in \mathcal{Q}_j^r$ such that $|Q| > \eta\lambda_r/(\eta - \underline{\ell})$. And write $T = Q \cap B_r(i)$, $F = Q \setminus B_r(i)$.

Case 1: $|F| \leq L_r$. Consider a transition link $e = T \to Q$ in the directed graph $\mathcal{T}_r$. Note that its weight is $w_r(e) = ((\eta - \underline{\ell})/\eta)|F|$. Hence by the definition of A_r, we get

$$A_r(Q) \geq A_r(T) + \frac{\eta - \underline{\ell}}{\eta}|F|. \tag{3}$$

The violation of the threshold inequality on Q requires

$$S(Q) = S(T) + S(F) < \eta(|T| + |F| - \lambda_r - A_r(Q)).$$

Since $T \in \mathcal{Q}_i^r$, the threshold inequality holds on it,

$$S(T) \geq \eta(|T| - \lambda_r - A_r(T)).$$

Combining the two inequalities with (3) gives:

$$S(F) < \eta|F| - \eta(A_r(Q) - A_r(T)) \leq \eta|F| - (\eta - \underline{\ell})|F| = \underline{\ell}|F|,$$

which is impossible because $S(F) \geq \underline{\ell}|F|$. Thus, small F's can never cause a failure.

Case 2: $|F| > L_r$. In this case, $w_r(e) = 0$. Hence, $A_r(Q) \geq A_r(T)$. By a similar argument as above, if the threshold inequality for Q fails, then

$$S(F) < \eta|F| - \eta(A_r(Q) - A_r(T)) \leq \eta|F|.$$

Conditional on $\theta = H$, the log-likelihood ratios in F are independent and have expectation κ. Since $\eta < \kappa$, by the definition of $c = 2(\kappa - \eta)^2/(\bar{\ell} - \underline{\ell})^2$ and Hoeffding inequality gives

$$\mathbb{P}\left(S(F) < \eta|F| \mid \theta = H\right) \leq \exp(-c|F|) \leq \exp(-cL_r).$$

Hence, by a union bound, we get

$$\begin{aligned}
\mathbb{P}(I_j^r \notin Y_j^r | I_i^r = x, \theta = H) &\leq \sum_{Q \in \mathcal{Q}_j^r} \mathbb{P}\left(S(Q) < \eta(|Q| - \lambda_r - A_r(Q)) \mid I_i^r = x, \theta = H\right) \\
&= \sum_{\substack{Q \in \mathcal{Q}_j^r: \\ |Q| > \eta\lambda_r/(\eta - \underline{\ell}), \\ |Q \setminus B_r(i)| > L_r}} \mathbb{P}\left(S(Q \setminus B_r(i)) < \eta|Q \setminus B_r(i)| \mid \theta = H\right) \\
&\leq |\mathcal{Q}_j^r| \exp(-cL_r).
\end{aligned}$$

Finally, taking a union bound over $j \in N_i$,

$$\mathbb{P}(I_j^r \notin Y_j^r \text{ for some } j \in N_i | I_i^r = x, \theta = H) \leq \sum_{j \in N_i} |\mathcal{Q}_j^r| \exp\left(-cL_r\right) \xrightarrow[r \to \infty]{} 0$$

uniformly over i, which follows from Condition (C.2).

□

Claim A.3.

$$\lim_{r \to \infty} \inf_{i \in N} \mathbb{P}(I_i^r \in Y_i^r | \theta = H) = 1.$$

Proof of Claim A.3. By a union bound,

$$\mathbb{P}(I_i^r \notin Y_i^r | \theta = H) \leq \sum_{Q \in \mathcal{Q}_i^r} \mathbb{P}\left(S(Q) < \eta(|Q| - \lambda_r - A_r(Q)) | \theta = H\right).$$

As shown in Claim A.2, Q may violate the threshold inequality only if $|Q| > \eta\lambda_r/(\eta - \underline{\ell})$, hence

$$\begin{aligned}
\mathbb{P}(I_i^r \notin Y_i^r | \theta = H) &\leq \sum_{\substack{Q \in \mathcal{Q}_i^r: \\ |Q| > \eta\lambda_r/(\eta - \underline{\ell})}} \mathbb{P}\left(S(Q) < \eta(|Q| - \lambda_r - A_r(Q)) | \theta = H\right) \\
&\leq \sum_{\substack{Q \in \mathcal{Q}_i^r: \\ |Q| > \eta\lambda_r/(\eta - \underline{\ell})}} \mathbb{P}\left(S(Q) < \eta|Q| \mid \theta = H\right) \leq |\mathcal{Q}_i^r| \exp(-c|Q|) \\
&\leq |\mathcal{Q}_i^r| \exp\left(-c\frac{\eta\lambda_r}{\eta - \underline{\ell}}\right) \xrightarrow[r \to \infty]{} 0,
\end{aligned}$$

where convergence holds uniformly over i and follows from Condition (C.3).

□

Lemma A.1. *Consider a family $(Y_i^r)_{i\in N}$ which is self-sustaining, i.e. for every i,*

$$Y_i^r \subseteq \mathcal{B}_{i,r}^q \left(H \cap \left(\bigcap_{j\in N_i} Y_j^r \right) \right).$$

Then, $Y_i^r \subseteq C_{i,r}^q$ for every i.

Proof of Lemma A.1. Similar to the proof of Proposition 1, we will prove by induction that for every agent i, $Y_i^r \subseteq C_{i,r}^{q,n}$ for every n and thus $Y_i^r \subseteq \bigcap_{n\geq 0} C_{i,r}^{q,n} = C_{i,r}^q$.

For $n = 0$, $Y_i^r \subseteq \Omega = C_{i,r}^{q,0}$. Now suppose that for every agent j, $Y_j^r \subseteq C_{j,r}^{q,k}$ for $k \leq (n-1)$. Then,

$$H \cap \left(\bigcap_{j\in N_i} Y_j^r \right) \subseteq H \cap \left(\bigcap_{j\in N_i} C_{j,r}^{q,n-1} \right).$$

Since the belief operator is monotone and $(Y_i^r)_{i\in N}$ are self-sustaining,

$$Y_i^r \subseteq \mathcal{B}_{i,r}^q \left(H \cap \left(\bigcap_{j\in N_i} Y_j^r \right) \right) \subseteq \mathcal{B}_{i,r}^q \left(H \cap \left(\bigcap_{j\in N_i} C_{j,r}^{q,n-1} \right) \right) = C_{i,r}^{q,n}.$$

It follows that $Y_i^r \subseteq C_{i,r}^{q,n}$ for every n and hence $Y_i^r \subseteq C_{i,r}^q$. □

Proof of Theorem 5. Fix any $q \in (0,1)$. We use the family $(Y_i^r)_{i\in N}$ constructed above to establish network q-learning. First, we show that this family is self-sustaining. Using the identity

$$\mathbb{P}(H, I_j^r \in Y_j^r \ \forall j \in N_i | I_i^r = x) = \mathbb{P}(H | I_i^r = x)\mathbb{P}(I_j^r \in Y_j^r \ \forall j \in N_i | I_i^r = x, \theta = H)$$

and Claims A.1 and A.2, we obtain

$$\lim_{r\to\infty} \inf_{i\in N} \inf_{x\in Y_i^r} \mathbb{P}(H,\ I_j^r \in Y_j^r \ \forall j \in N_i | I_i^r = x) = 1.$$

Since $q < 1$, it follows that, for all sufficiently large r for all i,

$$Y_i^r \subseteq \mathcal{B}_{i,r}^q \left(H \cap \left(\bigcap_{j\in N_i} Y_j^r \right) \right),$$

meaning that the family $(Y_i^r)_{i\in N}$ is self-sustaining. By Lemma A.1, $Y_i^r \subseteq C_{i,r}^q$. Therefore, by Claim A.3,

$$\inf_{i\in N} \mathbb{P}(C_{i,r}^q | \theta = H) \geq \inf_{i\in N} \mathbb{P}(I_i^r \in Y_i^r | \theta = H) \xrightarrow[r\to\infty]{} 1.$$

Hence, network q-learning of H is obtained. Since $q \in (0,1)$ was arbitrary, network common learning of H occurs. □

A.5 Proof of Theorem 2

We first establish in two lemmas two properties of convex sub-graphs of $\mathbb{Z}^k$ that will be used to verify conditions of Theorem 5. The first lemma bounds the number of test regions.

Lemma A.2. *If G is a convex sub-graph of $\mathbb{Z}^k$, then, for every i and r,*

$$|\mathcal{Q}_i^r| \le (2r+1)^{2^k}.$$

Proof of Lemma A.2. Since G that is a convex sub-graph of $\mathbb{Z}^k$, its graph distance admits the following representation

$$d(i,j) = \|i-j\|_1 = |i_1 - j_1| + \ldots + |i_k - j_k| = \max_{\epsilon \in \{\pm 1\}^k} \epsilon \cdot (i-j),$$

where $\cdot$ denotes scalar multiplication. Therefore,

$$B_r(i) = \{j \in N : \epsilon \cdot (j-i) \le r,\ \forall \epsilon \in \{-1,1\}^k\}.$$

Let $Q \in \mathcal{Q}_i^r$. Then $Q = \bigcap_{x \in C} B_r(x)$ for some finite connected C, containing i. Therefore,

$$Q = \left\{ j \in N : \epsilon \cdot j \le \min_{x \in C} \epsilon \cdot x + r,\ \forall \epsilon \in \{-1,1\}^k \right\}.$$

It means that Q is determined by 2^k integer values $\min_{x \in C} \epsilon \cdot x + r$ with $\epsilon \in \{-1,1\}^k$. Since $Q \subseteq B_r(i)$,

$$\epsilon \cdot i - r \le \min_{x \in C} \epsilon \cdot x + r \le \epsilon \cdot i + r,$$

and there can be at most $2r+1$ different values. Finally, there are 2^k choices of ϵ, hence $|\mathcal{Q}_i^r| \le (2r+1)^{2^k}$. □

The second lemma bounds the allowance term.

Lemma A.3. *If G is a convex sub-graph of $\mathbb{Z}^k$, then*

$$\sup_{Q \in \mathcal{Q}^r} A_r(Q) \le 2^{k+1} \frac{(\eta - \underline{\ell})}{\eta} r L_r.$$

Proof of Lemma A.3. We fix the final region $Q = Q_1$ an consider any directed path ending at it,

$$\gamma = Q_\tau \to Q_{\tau-1} \to \cdots Q_1 = Q.$$

Choose i such that $Q \in \mathcal{Q}_i^r$ and choose a finite set C_1 containing i such that $Q = \cap_{j \in C_1} B_r(j)$. For every t, choose an agent i_{t+1} witnessing the transition $Q_{t+1} \to Q_t$, so that

$$Q_{t+1} = Q_t \cap B_r(i_{t+1}), \quad F_{t+1} = Q_t \setminus B_r(i_{t+1})$$

Write $Q_t = \bigcap_{j \in C_t} B_r(j)$, where $C_{t+1} = C_t \cup \{i_{t+1}\}$.

For every $\epsilon \in \{-1,1\}^k$, the corresponding value $\min_{j\in C_t} \epsilon \cdot j + r$ is non-increasing in t. Moreover, if $F_{t+1} \neq \emptyset$, then there exists $x \in Q_t$, such that $x \notin B_r(i_{t+1})$. Thus, for some $\epsilon \in \{-1,1\}^k$,

$$\epsilon \cdot x > \epsilon \cdot i_{t+1} + r.$$

Moreover, since $x \in Q_t$, then

$$\epsilon \cdot x \leq \min_{j\in C_t} \epsilon \cdot j + r.$$

Combining two inequalities, we get,

$$\min_{j\in C_t} \epsilon \cdot j > \epsilon \cdot i_{t+1}.$$

So, the corresponding value for Q_{t+1}, i.e. $\min_{j\in C_t\cup\{i_{t+1}\}} \epsilon \cdot j + r = \epsilon \cdot i_{t+1} + r$ is smaller than the value for Q_t. Since values are integers, it decreases by at least 1.

Moreover, these values satisfy

$$\epsilon \cdot i - r \leq \min_{j\in C_t} \epsilon \cdot j + r \leq \epsilon \cdot i + r,$$

where the first inequality holds because $Q_t \subseteq Q_1 = Q \subseteq B_r(i)$, and thus for any $x \in Q_t$ we have $\epsilon \cdot i - r \leq \epsilon \cdot x \leq \min_{j\in C_t} \epsilon \cdot j + r$. And the second inequality follows from the fact that $i \in C_t$ and thus $\min_{j\in C_t} \epsilon \cdot j \leq \epsilon \cdot i$.

Thus, the corresponding value can decrease at most $2r$ times along the path for each ϵ. Since there are 2^k choices of ϵ, a path has at most $2^{k+1}r$ edges with $F_e \neq \emptyset$. Moreover, $w_r(e) = 0$ whenever $F_e = \emptyset$, while otherwise $w_r(e) \leq ((\eta - \underline{\ell})/\eta)L_r$, hence

$$A_r(Q) \leq 2^{k+1}\frac{\eta - \underline{\ell}}{\eta} r L_r.$$

□

Now we are ready to verify Conditions (C.1)-(C.3). Choose $L_r = C \log r$ and $\lambda_r = r \log r$, where $C > 0$ and will be chosen sufficiently large.

For Condition (C.1), by Lemma A.3, $A_r(B_r(i)) = O(r \log r)$. Thus,

$$\inf_{i\in N}(|B_r(i)| - \lambda_r - A_r(B_r(i))) = \inf_{i\in N}(|B_r(i)| - O(r \log r)) \xrightarrow[r\to\infty]{} \infty,$$

which follows from the theorem assumption $\lim_{r\to\infty} \sup_{i\in N} \frac{r \log r}{|B_r(i)|} = 0$.

For condition (C.2), by Lemma A.2 and the fact that every agent in the convex sub-graph of $\mathbb{Z}^k$ has at most $2k$ agents, we get

$$\sup_{i\in N} \sum_{j\in N_i} |\mathcal{Q}_j^r| \exp(-cL_r) = O(r^{2k} \exp(-cC \log r)) = O(r^{2k-cC}) \xrightarrow[r\to\infty]{} 0,$$

if we choose a sufficiently large constant C, such that $cC > 2^k$.

Finally, for condition (C.3), again by Lemma A.2, we get

$$\sup_{i\in N} |\mathcal{Q}_i^r| \exp\left(-\frac{c\eta}{\eta - \underline{\ell}}\lambda_r\right) = O\left(r^{2^k} \exp\left(-\frac{c\eta}{\eta - \underline{\ell}} r \log r\right)\right) = O\left(r^{2^k - \frac{c\eta}{\eta-\underline{\ell}} r}\right) \xrightarrow[r\to\infty]{} 0.$$

Thus, Conditions (C.1)-(C.3) are satisfied and the result follows from Theorem 5.

A.6 Proof of Theorem 3

For trees, we first introduce a convenient representation of test regions. In addition to balls centered at vertices, we allow a ball to also be centered at the midpoint of an edge. We need this enlargement because test regions are defined as intersections of balls, and on a tree the class of generalized balls is closed under such intersections, while the class of ordinary vertex-centered balls is not.

For an edge (i, j), call the midpoint as m, the distance from any point k to that midpoint is:

$$d(k, m) = 1/2 + \min\{d(k, i), d(k, j)\}.$$

For such a vertex or a midpoint k and for $s \in \frac{1}{2}\mathbb{N}$, define a generalized ball,

$$\widetilde{B}_r(i) = \{j \in N : d(j, i) \leq s\}.$$

Lemma A.4. *Every test region $Q \in \mathcal{Q}_i^r$ is a generalized ball.*

Proof of Lemma A.4. Let $Q \in \mathcal{Q}_i^r$. By definition, there is a finite connected set C with $i \in C$ such that $Q = \cap_{x \in C} B_r(x)$. Let $D = \max_{i,j \in C} d(i, j)$. Choose $i, j \in C$ such that $d(i, j) = D$ and let m be the midpoint of the unique path connecting i and j. For every $k \in N$,

$$\max_{t \in C} d(k, t) = d(k, m) + \frac{D}{2}.$$

Indeed, by triangle inequality and since $d(m, t) \leq D/2$ for every $t \in C$,

$$d(k, t) \leq d(k, m) + d(m, t) \leq d(k, m) + D/2.$$

Moreover, for i and j,

$$\max\{d(k, i), d(k, j)\} = d(k, m) + D/2,$$

hence the equality follows. Therefore,

$$k \in Q = \cap_{x \in C} B_r(x) \iff \max_{x \in C} d(k, x) \leq r \iff d(k, m) \leq r - \frac{D}{2},$$

implying

$$Q = \widetilde{B}_{(r - D/2)}(m)$$

if $D \leq 2r$ and is empty otherwise. □

The generalized-ball representation also allows us to bound the allowance term by controlling how many edges along a path can have positive weights. The following lemma gives the required bound.

Lemma A.5. *For every r,*

$$\sup_{Q \in \mathcal{Q}^r} A_r(Q) \leq 2 \frac{\eta - \underline{\ell}}{\eta} r L_r.$$

Proof of Lemma A.5. Consider an edge $T \to Q$ and two neighboring agents i, j witnessing this transition, with $Q = \tilde{B}_s(k) \in \mathcal{Q}_j^r$, with $T = Q \cap B_r(i)$ and $F = Q \setminus B_r(i)$. Suppose $F \neq \emptyset$, then there exists $x \in B_s(k)$ such that $x \notin B_r(i)$. Hence, $d(x,i) > r$. Since $d(x,k) \leq s$, we have

$$d(x,k) + d(k,i) \geq d(x,i) > r \implies d(k,i) > r - d(x,k) \geq r - s.$$

Furthermore, by Lemma A.4, T is a generalized ball. Its radius is

$$s' = \frac{r + s - d(i,k)}{2} < \frac{r+s}{2} - \frac{r-s}{2} = s \Rightarrow s - s' \geq 1/2.$$

Hence if $F \neq \emptyset$, then the intersection T is a generalized ball with radius reduced by at least $1/2$. Hence, along any directed path in $\mathcal{T}_r$, there can be at most $2r$ edges of positive weights. Moreover, by definition, $w_r(e) \leq (\eta - \underline{\ell})L_r$. It follows that $A_r(i,Q) \leq 2r((\eta - \underline{\ell})/\eta)L_r$. □

We now use the two lemmas to verify conditions (C.1)-(C.3). Choose $\lambda_r = C_\lambda r$ and $L_r = C_L r$ with $C_\lambda, C_L > 0$, which will be chosen sufficiently large. Let $d > 2$ be an upper bound on the degree of G. Since every test region is a generalized ball, we can bound the number of test regions by considering its every possible center and radius. The center must lie within distance r of i, giving $O((d-1)^r)$ possible centers, and for each center there are at most $2r+1$ possible half-integer radii. Hence,

$$|\mathcal{Q}_i^r| = O(r(d-1)^r).$$

For condition (C.1), by Lemma A.5,

$$\begin{aligned}\inf_{i \in N}(|B_r(i)| - \lambda_r - A_r(B_r(i))) &\geq \inf_{i \in N}\left(|B_r(i)| - C_\lambda r - 2\frac{\eta - \underline{\ell}}{\eta} C_L r^2\right) \\ &= \inf_{i \in N}\left(|B_r(i)| - O(r^2)\right) \xrightarrow[r\to\infty]{} \infty,\end{aligned}$$

which follows from the theorem assumption $\lim_{r\to\infty} \sup_{i \in N} r^2/|B_r(i)| \to 0$.

For Condition (C.2),

$$\begin{aligned}\sup_{i \in N} \sum_{j \in N_i} |\mathcal{Q}_j^r| \exp(-cL_r) &= O(r(d-1)^r \exp(-r \cdot cC_L)) \\ &= O\left(r \exp\left(r \log(d-1) - r \cdot cC_L\right)\right) \xrightarrow[r\to\infty]{} 0,\end{aligned}$$

if we choose C_L sufficiently large, such that $cC_L > \log(d-1)$.

For Condition (C.3),

$$\sup_{i \in N} |\mathcal{Q}_i^r| \exp\left(-\frac{c\eta}{\eta - \underline{\ell}}\lambda_r\right) = O\left(r \exp\left(r \log(d-1) - r\frac{c\eta C_\lambda}{\eta - \underline{\ell}}\right)\right) \xrightarrow[r\to\infty]{} 0,$$

if we choose C_λ sufficiently large, such that $c\eta C_\lambda/(\eta - \underline{\ell}) > \log(d-1)$.

Thus, Conditions (C.1)-(C.3) are satisfied and the result follows from Theorem 5.

A.7 Proof of Theorem 4

Similarly, to prove Theorem 4 we need to verify conditions (C.1)-(C.3). Suppose G has the similar neighborhood sizes property with constant K, the large information boundaries property, and subexponential-$\mathcal{Q}$. Choose $\varepsilon < \min\{\kappa, 1/(2K)\}$. Denote $\eta = \kappa - \varepsilon$. Also, choose $\lambda_r = \varepsilon \sup_{j\in N} |B_r(j)|$.

For condition (C.1), we first show that allowance is small. Recall that $w_r(e) = ((\eta - \underline{\ell})/\eta)|F|\mathbf{1}\{|F| \le L_r\}$. Consider a transition with a positive weight $T \to Q$, and choose neighbors i, j witnessing it. Suppose by contradiction,

$$|Q| \ge \frac{\varepsilon\eta|B_r(j)|}{\eta - \underline{\ell}} = \varepsilon'|B_r(j)|,$$

then by the large information boundaries condition with respect to ε', $F = Q \setminus B_r(i)$ is either empty or of size at least δr. Choose $L_r = \delta r/2$. Since the transition has a positive weight, then $F \neq \emptyset$ and $|F| \le L_r < \delta r$, which contradicts to the condition of large information boundaries. Thus, $|Q| < \varepsilon' \sup_{j\in N} |B_r(j)|$.

Along a directed path $\gamma = Q_\tau \to \cdots \to Q_1$, the test regions are nested $Q_\tau \subseteq \cdots \subseteq Q_1$ and the sets $F_{e_t} = Q_t \setminus Q_{t+1}$ are pairwise disjoint. Hence,

$$\sum_{e\in\gamma} w_r(e) \le \frac{\eta - \underline{\ell}}{\eta} \sum_{e: w_r(e)>0} |F_e| \le \frac{\eta - \underline{\ell}}{\eta}\varepsilon' \sup_{j\in N} |B_r(j)|.$$

Taking the supremum over paths $\gamma \to Q$,

$$A_r(Q) \le \frac{\eta - \underline{\ell}}{\eta}\varepsilon' \sup_{j\in N} |B_r(j)| = \varepsilon \sup_{j\in N} |B_r(j)| = \lambda_r.$$

Recall from the definition of similar neighborhood sizes, there exists a constant K such that for all i, j and r we have $|B_r(i)|/|B_r(j)| \le K$. In particular, $K \ge \sup_{j\in N} |B_r(j)|/|B_r(i)|$. Since $A_r(B_r(i)) \le \lambda_r = \varepsilon \sup_{j\in N} |B_r(j)|$, we get

$$\begin{aligned} |B_r(i)| - \lambda_r - A_r(B_r(i)) &= |B_r(i)| - 2\varepsilon \sup_{j\in N} |B_r(j)| \\ &\ge (1 - 2K\varepsilon)|B_r(i)| \\ &\ge (1 - 2K\varepsilon)(r+1) \xrightarrow[r\to\infty]{} \infty. \end{aligned}$$

The positivity of the coefficient is guaranteed by the choice of $\varepsilon < \min\{\kappa, 1/(2K)\}$.

For condition (C.2), the number of i's neighbors is uniformly bounded over i, by similar neighborhood sizes for $r = 1$. Thus, there exist a constant $D < \infty$ such that

$$\begin{aligned} \sup_{i\in N} \sum_{j\in N_i} |\mathcal{Q}_j^r| \exp(-cL_r) &\le D \sup_{j\in N} |\mathcal{Q}_j^r| \exp(-cL_r) \\ &= D \sup_{j\in N} |\mathcal{Q}_j^r| \exp(-c\delta r/2) \xrightarrow[r\to\infty]{} 0, \end{aligned}$$

where convergence follows from the subexponential-$\mathcal{Q}$ condition.

For condition (C.3),

$$\begin{aligned}\sup_{i\in N}|\mathcal{Q}_i^r|\exp\left(-\frac{c\eta}{\eta-\underline{\ell}}\lambda_r\right) &= \sup_{i\in N}|\mathcal{Q}_i^r|\exp\left(-\frac{c\eta\varepsilon}{\eta-\underline{\ell}}\lambda_r\sup_{j\in N}|B_r(j)|\right)\\ &\leq \sup_{i\in N}|\mathcal{Q}_i^r|\exp\left(-\frac{c\eta\varepsilon}{\eta-\underline{\ell}}\lambda_r(r+1)\right)\xrightarrow[r\to\infty]{} 0,\end{aligned}$$

where convergence follows from the subexponential-$\mathcal{Q}$ condition.

Thus, Conditions (C.1)-(C.3) are satisfied and the result follows from Theorem 5.